Kelp Density Relative to the Preceding Five Years

Following the methods of Frieder et al. (2025), we express trends in understory kelp density relative to the preceding five years as follows.[1]

For species s, site i, and current year t, the current density is defined as

$$D_{s,i,t}^{\text{current}} = D_{s,i,t}, \tag{1}$$

where $D_{s,i,t}$ is the observed density of species s at site i during the current year t.
The mean density during the five preceding years is defined as

$$D_{s,i,t}^{5\text{yr}} = \frac{1}{5}\sum_{k=1}^{5} D_{s,i,t-k}. \tag{2}$$

We define the current kelp-density metric as the current density divided by the mean density during the five preceding years, multiplied by 100:

$$K_{s,i,t} = 100\left(\frac{D_{s,i,t}}{\frac{1}{5}\sum_{k=1}^{5} D_{s,i,t-k}}\right). \tag{3}$$

Here,

$$\begin{aligned} K_{s,i,t} &= \text{current density as a percentage of the preceding} \\ &\quad\ \text{five-year mean}, \\ D_{s,i,t} &= \text{density of species } s \text{ at site } i \text{ in year } t, \\ D_{s,i,t-k} &= \text{density of species } s \text{ at site } i,\ k \text{ years before } t. \end{aligned}$$

The resulting metric may be interpreted as follows:

- $K_{s,i,t} = 100$: current density equals the preceding five-year mean;
- $K_{s,i,t} > 100$: current density is greater than the preceding five-year mean;
- $K_{s,i,t} < 100$: current density is lower than the preceding five-year mean.

For example, $K_{s,i,t} = 75$ indicates that current density is 75% of the preceding five-year mean, whereas $K_{s,i,t} = 140$ indicates that current density is 140% of the preceding five-year mean.

This example expresses the metric at the site scale. Site-level metrics can then be averaged to generate a basin-scale metric, and basin-scale metrics can likewise be averaged to produce an overarching Puget Sound–wide metric.

[1] Frieder, C. A., Bell, T. W., Berry, H., Cavanaugh, K., Claar, D. C., Freiwald, J., Grime, B., Hamilton, S., Houskeeper, H. F., Lombardo, N., Marion, S., McHugh, T. A., McKenna, G., Parnell, P. E., Spector, P., and Weisberg, S. B. (2025). Developing a Status and Trends Assessment for Floating Kelp Canopies across Large Geographic Areas. *Environmental Science & Technology*. `https://doi.org/10.1021/acs.est.5c07501`.